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MIL-S-71146 (AR)
21 June 1993

MILITARY SPECIFICATION

Simulator, Launching, Anti-Tank, Guided Missile and Rocket M22 Parts for and Loading, Assembling and packing

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts for and loading, assembling and packing for one type of simulator designated as Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1370

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STANDARDS

MIL-STD-105	Sampling procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1234 -	Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1235 -	Single and Multi-Level Continous Sampling Procedure and Tables For Inspection By Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

Product Drawings

12953563 -	Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22

Packaging Drawings

9344296 -	Packing and Marking for Box, Packing Wood Wire Bound, for Box Fiberboard, for Simulator, Launching, AT, GM and RKT, M22
9344297 -	Packing and Marking for Box, Fiberboard, for Simulator, Launching, AT, GM and RKT, M22

Inspection Equipment Drawings

11749562 -	Simulator, Antitank Missile Fire

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings, specifications and standards.

3.2 Assembly. The Simulator Assembly shall comply with all requirements specified on dwg. 12953563, and all associated drawings, and with all requirements specified in applicable specifications and standards.

3.2.1 Contact closure switch. The output end of the cartridge shall contain an electrical contact strip (contact closure switch, Part Number 11749641). This strip shall have less than 1 ohm resistance when measured as specified in 4.5.5.

3.3. Moisture and volatile content.

3.3.1 Flash and bang. The moisture and volatile content of these compositions at the time of loading shall not exceed 0.15 percent.

3.3.2 Paper and paper products. The moisture and volatile content of the paper and paper products components of the simulator assembly at the loading station at the time of loading shall not exceed 6.0 percent.

3.4 Functioning. (Ambient Temperature 21±3°C, 70±5°F, Hot 51.5±3°C, 125±5°F, Cold -32±3°C, -25±5°F). The assembly shall function and comply with the following requirements after conditioning at specified temperature.

3.4.1 Sound characteristics. The sound level intensity of the cartridge functioning at a distance of 64 ± 1.0 feet (ft.) from the simulator assembly shall be 134 +6 - 7 decibles (db) for a total duration of 15 milliseconds (ins) maximum (max), with a rise-time of 5 ms max.

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3.4.2 Functioning time. The time from application of nine plus or minus one (9 + 1) volts direct current (VDC), with no current limitations, to audible report of 3.4.1 of the assembly shall not be greater than 0.110 seconds.

3.4.3 Flash characteristics. The cartridge shall produce a visible flash when fired from a launch tube. This flash shall be a minimum (rein) of six (6) feet in length, two (2) feet in width with a duration of 40 to 500 ms in winds of not more than 8 miles per hour. Residual smoke shall be white in appearance.

3.5 Leak test. The cartridge shall show no evidence of air or water leakage when tested as specified in 4.5.6.

3.6 Jumble. The simulator shall comply with the following requirements.

a. The simulator shall not function during test.

b. There shall be no loose pyrotechnic powder on the simulator or in the test box after jumbling.

c. The simulator components shall not separate during test.

3.7 First article. When specified in the contract or purchase order (see 6.3), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.8 Workmanship All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this . specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection. (see 4.3)

b. Quality Conformance Inspection. (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated:

Part Description	Drawings	Quantity
Case	11749631	5/cavity
Tube	11749635	25
Closure, Switch	11749641	10
Closure	11749638	5/cavity
Disc, Separator	11749637	25
Pellet Assembly	11749633	5
End Plate	11749632	5/cavity
Packing & Marking	9344297	1 box/150
Simulator, Launching, Anti- Tank, Guided Missile and Rocket M22	12953563	175

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22 and Components	SHEET 1 OF 3	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Case (Dwg. 11749631) Examination for defects	5/mold cavity	3.2	4.4.2.1 4.5.1
	Tube (Dwg. 11749635) Examination for defects Moisture content	25 (a)	3.2 3.3.2	4.4.2.2 4.5.2.2
	Closure Switch (Dwg. 11749641) Examination for defects Resistance	10 (a)	3.2 3.2.1	4.4.2.3 4.5.5
	Closure (Dwg. 11749638) Examination for defects	5/mold cavity	3.2	4.4.2.4 4.5.1
	Disc, Separator (Dwg. 11749637) Examination for defects	25	3.2	4.4.2.5

NOTES:
(a) Sample for this test shall be taken from test samples.

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TABLE I First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22 and Components	SHEET 2 OF 3		DRAWING NUMBER See below
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Pellet Assembly (Dwg. 11749633) Examination for defects	5	3.2	4.4.2.6
	End Plate (Dwg. 11749632) Examination for defects	5/mold cavity	3.2	4.4.2.7 4.5.1
	Packing and marking (Dwg. 9344297) Examination for defects	15 subpacks	3.2	4.4.2.10-4.4.2.12
	Vacuum chamber leak test	(b)	3.5	4.5.8
	Simulator, Launching, Guided Missile and Rocket, M22 (Dwg. 12953563) Examination for defects	175	3.2	4.4.2.8, 4.4.2.9
	Leak test	175	3.5	4.5.6
	Functioning (Cold temperature)	50	3.4	4.5.3
	Sound characteristics		3.4.1	4.5.3.1
	Function time		3.4.2	4.5.3.2
	Flash characteristics		3.4.3	4.5.3.3

NOTES:
(b) Leak test to be performed on above units.

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TABLE I First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22 and Components	SHEET 3 OF 3	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Functioning (Hot temperature)	50	4.3	4.5.3
	Sound characteristics		4.3.1	4.5.3.1
	Function time		3.4.2	4.5.3.2
	Flash characteristics		3.4.3	4.5.3.3
	Functioning (Ambient temperature)	50	3.4	4.5.3
	Sound characteristics		3.4.1	4.5.3.1
	Function		3.4.2	4.5.3.2
	Flash characteristics		3.4.3	4.5.3.3
	Jumble	25	3.6	4.5.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Simulator Assemblies shall contain:

a. Parts of one kind from one manufacturer.

b. Percussion Primer from not more than one lot interfix number from one manufacturer.

c. Composition ingredients from not more than one lot or batch (applicable only when not manufactured by the loading contractor).

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. Where cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II Attributes sampling inspection.

Lot Size	Inspection Level I	II	III	IV
2 to 8	*	*	*	5
9 to 15	*	*	13	5
16 to 25	*	*	13	5
26 to 50	*	32	13	5
51 to 90	*	32	13	13
91 to 150	125	32	13	13
151 to 280	125	32	32	20
281 to 500	125	32	32	20
501 to 1200	125	80	50	20
1201 to 3200	125	80	50	32
3201 to 10000	125	125	50	32
10001 to 35000	315	125	80	50
35001 to 150000	315	125	80	50
150001 to 500000	500	200	125	50
500001 and over	500	200	125	50

Numbers under inspection levels indicate sample size; asterisks indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.2). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH 4.4.2.1	TITLE Case	SHEET 1 OF 1	DRAWING NUMBER 11749631
			NEXT HIGHER ASSEMBLY 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	End plate seat depth to bottom groove	(a)	3.2	Gage
102	End plate groove width	(a)	3.2	Gage
103	Inside length	(a)	3.2	Gage
104	Closure switch counterbore diameter	(a)	3.2	Gage
105	Closure seat depth	(a)	3.2	Gage
Minor				
201	Flange O.D.	(b)	3.2	Gage
202	Small O.D.	(b)	3.2	Gage
203	Poor workmanship	(c)	3.8	Visual

NOTES: See 4.5.1 for qualification and production control of mold cavities. (a) Table II, Level III, (b) Table II, Level III, (c) Table II, Level IV

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.2	TITLE Tube	SHEET 1 OF 1	DRAWING NUMBER 11749635
			NEXT HIGHER ASSEMBLY 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter, (min.)	(a)	3.2	Gage
102	Outside diameter, (max.)	(a)	3.2	Gage
Minor				
201	Total length	(b)	3.2	Gage
202	Poor workmanship	(c)	3.8	Visual

NOTES:
(a) Table II, Level II, (b) Table II, Level III, (c) Table II, Level IV

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Closure Switch	1 OF 1	11749641
			NEXT HIGHER ASSEMBLY 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter, (max.)	(b)	3.2	Gage
102	Resistance, (max.)	(a)	3.2	Gage
Minor				
201	Poor workmanship	(c)	3.8	Visual

NOTES: (a) Table II, Level I, (b) Table II, Level II, (c) Table II, Level IV

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Closure	1 OF 1	11749638
			NEXT HIGHER ASSEMBLY 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Flange width	(a)	3.2	Gage
102	Large Diameter	(a)	3.2	Gage
103	Small outside diameter	(a)	3.2	Gage
104	Groove diameter	(a)	3.2	Gage
Minor				
201	Poor workmanship	(b)	3.8	Visual

NOTES:
See 4.5.1 for qualification and production contol of mold cavities.
(a) Table II, Level II, (b) Table II, Level IV

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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PARAGRAPH 4.4.2.5	TITLE Disc, Separator	SHEET 1 OF 1	DRAWING NUMBER 11749637
			NEXT HIGHER ASSEMBLY 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Diameter	(a)	3.2	Gage
Minor 201	Poor workmanship	(b)	3.8	Visual

NOTES: (a) Table II, Level II, (b) Table II, Level IV

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Pellet Assembly	1 OF 1	11749633
			NEXT HIGHER ASSEMBLY: 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Flash composition above flush	(a)	3.2	Gage
102	First fire missing, less than 75% coverage	(a)	3.2	Visual
Minor				
201	Flash composition greater than specified dimension below flush	(b)	3.2	Gage
202	Poor workmanship	(c)	3.8	Visual

NOTES: (a) Table II, Level I, (b) Table II, Level III, (c) Table II, Level IV

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	End Plate	1 OF 1	11749632
			NEXT HIGHER ASSEMBLY: 12953563

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter (primer hole)	(a)	3.2	Gage
102	Large outside diameter	(a)	3.2	Gage
103	Small outside diameter	(a)	3.2	Gage
104	Depth of primer hole	(a)	3.2	Gage
105	Overall thickness	(a)	3.2	Gage
Minor				
201	Poor workmanship	(b)	3.8	Visual

NOTES:
See 4.5.1 for qualification and production control of mold cavities
(a) Table II, Level II, (b) Table II, Level IV

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Simulator, Launching, AT, GM and RKT M22 (prior to assy of Closure Charge Cover)	1 OF 1	12953563
			NEXT HIGHER ASSEMBLY ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Moisture content	(a)	3.2	4.5.2.1/Test
102	Weight of bang composition (verification)	50	3.2	Balance (Note c or d)
Minor				
201	Poor workmanship	(b)	3.8	Visual

NOTES: (a)See para. 4.4.3.1, (b) Table II, Level IV, (c) A sample of fifty(50) loading assemblies, prior to sealing the closure shall be selected from each eight(8) hours of production and subjected to a weight of charge verification as follows: Using a tared bang composition container, the weight difference due to the addition of the bang composition shall be determined. If the weight of any charge fails to meet the applicable drawing requirement, those assemblies loaded since the last acceptable verification test shall be classed defective and removed from the lot. (Non-destructive test)., (d) MIL-STD-1235 single and multi-level continuous sampling procedure may be used.

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PARAGRAPH 4.4.2.9	TITLE: Simulator, Launching, AT, GM and RKT M22	SHEET 1 OF 2		DRAWING NUMBER 12953563
				NEXT HIGHER ASSEMBLY ---
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical 1	Evidence of loose powder on exterior of assembly	100%	3.2	Visual
Major 101	Assembly damaged to extent that powder may leak	(c)	3.2	Visual
102	Primer above flush or greater than max below flush	(c)	3.2	Gage
103	Closure switch above flush or loose	(c)	3.2	Gage
104	Any component(s) loose at any joint or contact	(c)	3.2	Visual/Manual
105	Epoxy sealant missing, peeling or incorrectly applied	(c)	3.2	Visual
106	Closure switch conductive area separated or torn	(d)	3.2	Visual
107	Total length	(d)	3.2	Gage

NOTES: a) After three(3) consecutive lots have been accepted, sample may be reduced in accordance with para. 4.4.5. (b) After 3 consecutive lots have been acccepted, sound intensity, rise time and duration time testing shall be performed every tenth(10th) lot or the last lot less than ten. If a lot is rejected, test frequency shall revert to para. 4.4.5.

AMSMC Form 1570a, 1 Jul 89

Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Simulator, Launching, AT, GM and RKT M22	2 OF 2	12953563
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
108	Leak test	150(a),2-3	3.5	4.5.6
109	Functioning (Hot)	50 (a),2-3	3.4	4.5.3, 4.4.5.2
	Sound characteristics	(b)	3.4.1	4.5.3.1
	Functioning time		3.4.2	4.5.3.2
	Flash characteristics		3.4.3	4.5.3.3
110	Functioning (Cold)	50(a),3-4	3.4	4.5.3, 4.4.5.3
	Sound characteristics	(b)	3.4.1	4.5.3.1
	Functioning time		3.4.2	4.5.3.2
	Flash characteristics		3.4.3	4.5.3.3
111	Functioning (Ambient)	50(a),2-3	3.4	4.5.3, 4.4.5.2
	Sound characteristics	(b)	3.4.1	4.5.3.1
	Functioning time		3.4.2	4.5.3.2
	Flash characteristics		3.4.3	4.5.3.3
Minor				
201	Marking, missing, incorrect or illegible	(e)	3.2	Visual
202	Poor workmanship	(f)	3.8	Visual

NOTES: (c) Table II, Level I, (d) Table II, Level II, (e) Table II, Level III, (f) Table II, Level IV

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-71146 (AR)

PARAGRAPH 4.4.2.10	TITLE Packing and Marking for box, Fiberboard, Simulator, launching, AT, GM and RKT, M22, prior to sealing	SHEET 1 OF 1	DRAWING NUMBER 9344297
			NEXT HIGHER ASSEMBLY 9344296

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Loose fit of partition to box and orientation of cutouts incorrect	(b)	3.2	Manual
102	Verify primer end up and count	(a)	3.2	Visual
103	Loose pack	(b)	3.2	Visual
104	Filler sheet missing or coated surface not facing simulator	(b)	3.2	Visual
Minor				
201	Poor workmanship	(c)	3.8	Visual

NOTES (a) Table II, Level I, (b) Table II, Level II, (c) Table II, Level IV

AMSMC Form 1570, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-71146 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Packing and Marking for box, Fiberboard, Simulator, launching, AT, GM and RKT, M22, prior to packing in wirebound box)	1 OF 1	9344297
			NEXT HIGHER ASSEMBLY 9344296

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Heat seal incorrect on barrier bag	10, 0-1	3.2	4.5.7
102	Marking incorrect or not legible	(a)	3.2	Visual
Minor				
201	Barrier bag fold incorrect	(b)	3.2	4.5.7
202	Poor workmanship	(c)	3.8	Visual

NOTES: (a) Table II, Level II, (b) Table II, Level III, (c) Table II, Level IV

AMSMC Form 1570, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-71146 (AR)

PARAGRAPH 4.4.2.12	TITLE Packing and Marking for box, Packing, Wood, For Box, Fiberboard For Simulator Launching, AT, GM & RKT, M22	SHEET 1 OF 1	DRAWING NUMBER 9344296
			NEXT HIGHER ASSEMBLY ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Bottom, sides or end fillers missing	(a)	3.2	Visual
102	Incorrect number of fiberboard boxes	(a)	3.2	Visual
103	Top insert filler missing	(a)	3.2	Visual
104	Loose pack	(a)	3.2	Manual
Minor				
201	Marking incorrect or not legible	(b)	3.2	Visual
202	Poor workmanship	(c)	3.8	Visual

NOTES
(a) Table II, Level II, (b) Table II, Level III, (c) Table II, Level IV

AMSMC Form 1570, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used

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4.4.3 Testing.

4.4.3.1 Moisture content (see Table III) - Major Defects.

TABLE III

Flash and bang	See 3.3.1
Paper and paper products	See 3.3.2

The contractor shall provide adequate controls to insure that the material comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight(8) hours production. If a composite sample exceeds the requirement, that quantity or sub-lot material represented by the sample-shall not be used in loading and packing, the remaining unloaded and upacked material shall not be used in . production, and the loaded and packed simulators shall be rejected. Test shall be performed as specified in 4.5.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.5 Functioning test.

4.4.5.1 Leak test. Beginning with the first lot produced and continuing until three(3) consecutive lots have complied with the acceptance criteria of 4.4.2.9, one hundred and fifty (150) simulator assemblies shall be selected from each lot for test, after three (3) consecutive lots have met the acceptance criteria of 4.4.2.9, eighty (80) simulators shall be selected from each lot for this test. The lot shall be rejected if two (2) or more major functioning defects occur, and lot sampling shall revert to one hundred and fifty (150) samples per lot in accordance with 4.4.2.9.

4.4.5.2 Hot and ambient temperature. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria of 4.4.2.9, fifty (50) simulator assemblies shall be selected from each lot for test. After three (3) consecutive lots have met the acceptance criteria of 4.4.2.9, thirty-two (32) simulators shall be selected from each lot for this test. The lot shall be rejected if two (2) or more major functioning defects occur, and lot sampling shall revert to fifty (50) samples per lot in accordance with 4.4.2.9.

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4.4.5.3 Cold temperature. Beginning with the first lot produced and continuing until three (3) consecutive losts have complied with the acceptance criteria of 4.4.2.9, fifty (50) simulator assemblies shall be selected from each lot for test. After three (3) consecutive lots have met the acceptance criteria of 4.4.2.9, thirty-two (32) simulators shall be selected from each lot for this test. The lot shall be rejected if three (3) or more major functioning defects occur, and lot sampling shall revert to fifty (50) samples per lot in accordance with 4.4.2.9.

4.5 Methods of inspection.

4.5.1 Dimensional control of molded or plastic parts. In place of normal sampling and inspections associated with the classification of defects, and after a curing time for the parts has been approved (see note 1) a sample of three (3) Parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for a least the defects listed in 4.4.2.1, 4.4.2.4 and 4.4.2.7 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect. All cavities removed from production because of some fault, may, after reworking be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting Officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be submitted per 6.3 for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of the paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. reins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to the technical agency specified in the contract. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

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4.5.2 Moisture and volatile content.

4.5.2.1 Flash and bang. The percentage of moisture and volatile content in the flash composition shall be determined in accordance with Method 102.2.1 of MIL-STD-1234 except that the sample shall be heated until a constant weight is obtained in lieu of four hours. observation shall be made for compliance with the applicable requirement.

4.5.2.2 Paper and paper products. The moisture content shall be determined in accordance with Method 102.1.1 of Standard MIL-STD-1234 except that a ten (10) gram sample shall be used and the drying time shall be for two (2) hours. observation shall be made for compliance with the applicable requirement.

4.5.3 Functioning . (Ambient Temperature, 21 ± 3°C, 70 ± 5°F, Hot, 51.5 ± 3°F, 125°F ± 5°F Cold, -32 ± 3°C -25 ± 5°F). The simulator assemblies shall be conditioned for 12 hours min. at the specified temperature and fired within 5 minutes after removal from the conditioning chamber. The simulator shall be mounted in an approved fixture (simulating the Antitank, Missile Fire) on a rigid support and fired by means of an approved power source in accordance with 3.4.2 and firing device (P/N 11749562). (Destructive test)

4.5.3.1 Sound characteristics. Measurement of the sound intensity shall be taken at right angle to the front of the test fixture. The test item and sound detector shall be positioned on a common plane 6 ± .2 feet above ground. Observation shall be made for compliance with the applicable requirement.

4.5.3.2 Functioning time. The timing equipment shall be in accordance with 4.4.4. observation shall be made for compliance with the applicable requirement.

4.5.3.3 Flash characteristics. The size and duration of the flash/plume may be determined in accordance with standard practice. These methods can include, but not be limited to, visual or photographic processes using a grid background. Observation shall be made for compliance with the applicable requirement.

4.5.4 Jumble. The simulator assemblies shall be tested in accordance procedures specified in Standard MIL-STD-331 Test 102, except that the duration of the test shall be limited to a minimum of 20 minutes. The assemblies and equipment shall be observed and inspected for compliance with applicable requirements. (Destructive test)

4.5.5 Contact closure switch. The resistance of the closure switch shall be measured from edge to edge using any type of calibrated ohmmeter. The ohmmeter must be equipped with sharp pointed probes which can penetrate the protective coating on the contact switch. The test shall meet the requirements of 3.2.1. Parts tested shall not be returned to the lot. (Destructive test)

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4.5.6 Leak test. The cartridges shall be tested for leakage by submersing them in water (at 70° ± 5°F) contained in a vacuum chamber. '(Eliminate closure switch for leak test.) A vacuum differential of at least 6 inches of mercury shall be drawn and held for 30 sec. minimum while observing for air bubbles. A steady stream or recurring stream from any surface or seam shall constitute a failure. The cartridges shall be wiped dry prior to functioning. (Non-Destructive Test)

Alternative leak test. The cartridges shall be placed in a sealed test chamber and the chamber pressurized with 3 to 5 psi dry air and held for at least 15 seconds. Observe pressure gauge for pressure drop. Any cartridge that fails to comply with the applicable requirements shall consitute a failure.

4.5.7 Heat seal of barrier bag. The test shall be performed in accordance with the procedure specified in MIL-P-116, 4.4.4.3. After the test, the seal shall be examined for compliance with the applicable requirements (Destructive Test.)

4.5.8 Vacuum chamber leak test. The test shall be performed in accordance with the procedure specified in MIL-P-116, 4.4.3.2. Test samples shall be observed for compliance with the applicable requirements. (Non-Destructive Test)

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified on dwg. 9344296 and 9344297.

5.2 Packing and marking. Packing and marking shall be as specified on dwgs. 9344296 and 9344297.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the Simulator, Launching, Anti-Tank Guided Missile and Rocket, M22.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

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6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified-on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Distribution of ammunition data cards. Distribution of data cards include the foilOwing, Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-M, Picatinny Arsenal, NJ 07806-5000.

6.5 Humidity control. pyrotechnic compositions are hydroscopic materials and will deteriorate rapidly when exposed to moisture. It is highly desirable that all the ingredients of the simulator be carefully safe guarded from exposure to moisture and humid atmosphere. For assurance of the satisfactory loading of the simulator it is recommended that the blending and loading building be air conditioned in a manner that will maintain the relative humidity of the atmosphere below 60%.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.7 subject term (key word) listing.

Bang composition
Closure switch
End plate
Flash composition
Flash/Sound
Separator disc
Pellet assembly

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1370-A341)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-71146 (AR)	930621

3. DOCUMENT TITLE
SIMULATOR, LAUNCHING, ANTI-TANK, GUIDED MISSILE AND ROCKET M22 PARTS & LAP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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